

7	Answer: C
	$F = ma$ $F_{driving} - k(20)^2 = 1000(0.50)$ $F_{driving} = 500 + 400k$ $P = F_{driving} v$ $40 \times 10^3 = (500 + 400k)(20)$ $k = 3.75$ $P_{new} = F_{driving} v$ $P_{new} = F_{friction} v$ $P_{new} = kv^3 = (3.75)(25)^3 = 59 \text{ kW}$

8	Answer: C
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